

Deep learning for Coronary Artery Segmentation: CNN vs VIT

by

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Abstract

Cardiovascular diseases remain a leading cause of death worldwide, with coronary heart disease (CHD) being a significant contributor. It affects around 2.3 million people in the UK, and a key step to treating CHD is early detection and accurate assessment of coronary arteries, which is crucial for effective diagnosis and treatment planning. Currently, cardiologists are required to spend hours manually segmenting the coronary artery from CT scans per patient. This project aims to use deep learning techniques to create a 3D segmentation model that can automatically highlight the location of coronary arteries in cardiac CT scans, improving the efficiency and the accuracy of cardiac imaging, and speeding up the process of segmenting the artery. We set up multiple deep learning models with different architectures (detailed later), split between CNN and transformer models, and compared their efficacy and performance. We measured their capabilities mainly using the DICE metric, a measure of how the image segmented by the model compares to the actual image. The highest performing model was the DynUNet, with a dice score of 0.731, which was a CNN model, however, due to noise in the final segmented image it produced, it could be argued that SwinUNetR (0.720 dice), which was a vision transformer based model, was better, as the mask it produced was better. Overall, excluding anomalous results, both types of models achieved similar results, with CNNs at an average of 0.670, and vision transformers slightly higher at 0.687. This suggests that both are usable in this task, and there are many ways to improve their performance in future, such as by using more epochs.

Introduction

Cardiovascular disease is the number one cause of death worldwide¹. It kills almost 18 million people a year, accounting for about 15%² of all deaths in 2021, above even COVID-19 (the second highest)³. The most common type of cardiovascular disease is coronary artery disease, which is caused by a build-up of plaque in the coronary arteries. One method of finding this buildup of plaque is using Computed Tomography Angiography (CTA) scans, which work via firing beams of X-rays which are rotated around the body to generate 3D images in slices providing ‘non-invasive and... high-resolution 3D imaging’⁴. To quantify the level of plaque burden, and therefore the disease, doctors spend hours to manually separate the artery from the unnecessary portions of the image, then identify and inspect the plaque buildup area to form a final diagnosis/treatment plan. This work is rather monotonous and prone to error, and overall is time that ought to be spent on more productive matters, leading to the idea of automating it. This idea does come with challenges, such as differences in the bodies of people, for example the coronary arteries being located within muscle instead of a layer of fat, and noise in the CTA images. The structures of the arteries are also quite complex, again making it challenging to automate. To overcome these challenges, AI has been proposed as a tool to perform this segmentation. This has already been tried before, with many papers proposing various solutions. However, there has not been much study in the way of comparing the efficacy of these proposed methods, which is what we aim to do. Since deep learning models seem to outperform more traditional machine learning models, we have decided to focus on comparing these deep learning models more in our paper.

Aims

We aim to compare the efficacy of certain deep learning models, by primarily comparing their DICE similarity coefficients scores (what percentage of the predicted mask matches the actual mask), and other such metrics, in order to form a fair comparison between these models, as it is difficult to compare efficacy across papers, due to variations in things such as the datasets

¹World Health Organization (2025). *Cardiovascular diseases*. [online] World Health Organisation. Available at: https://www.who.int/health-topics/cardiovascular-diseases#tab=tab_1.

² World Health Organization (2024). *The top 10 causes of death*. [online] World Health Organization. Available at: <https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death>.

³World Health Organization (2024). *The top 10 causes of death*. [online] World Health Organization. Available at: <https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death>.

⁴ NIH (2022). *Computed Tomography (CT)*. [online] National Institute of Biomedical Imaging and Bioengineering. Available at: <https://www.nibib.nih.gov/science-education/science-topics/computed-tomography-ct>.

used. The size of previously publicly available datasets ([Kirişli H.A. et al.](#), [Schaap M. et al.](#)) were less than fifty images each, which is rather limited in scope. We used a dataset of 1000 3D cardiac CT scans ([ImageCAS](#)) but only took the first 100 to cut down on runtime and computation required, still vastly increasing the number of images used in most works.

Objectives

1. Research and understand the current state-of-the-art in deep learning-based medical image segmentation, with a focus on 3D cardiac CT scans.
2. Collect and preprocess a dataset of cardiac CT scans to be able to make fair comparisons between models.
3. Research and compare multiple deep learning models, split between Vision Transformer and CNN based architectures, using metrics such as the mean dice score.
4. Compare the models' segmentation results with manual annotations by radiologists to assess clinical relevance and accuracy.
5. Reach a conclusion about which models seem to perform best in this context.

Literature Review

The idea of using AI models to segment the coronary arteries has been around for a while, and as such, there is already a wealth of literature regarding it. Since we have decided to focus on comparing CNN and Transformer models, we have only reviewed the literature around the deep learning-based approaches, rather than the machine learning based approaches. One of the first to introduce the deep learning-based approach was [Moeskops et al., 2016](#), who investigated the use of a single CNN to perform segmentation, including in the context of coronary artery segmentation. This has been built on with many other advancements, such as the introduction of V-NET architecture, a 3D approach to segmentation (as opposed to CNN, which is better suited to 2D applications), with a Fully Convolutional Network (FCN) instead ([Milletari, F., Navab, N. and Ahmadi, S.-A. \(2016\).](#)), 3D multi-channel U-Net architecture ([Chen et al., 2019](#)), and transformer based models such as UNETR, where the encoder for the FCN is a transformer ([Hatamazideh et al., 2022](#)). (Any relevant models will be discussed in further detail below). In terms of comparing the efficacy of different models [Zeng et al., 2023](#) has started this process, comparing the optimal results of different methods of segmentation, along with uploading their own proprietary dataset of 1000 images. Their results are tabled below.

Method	Input type	Input size	Dice score (%)	HD (mm)	AHD (mm)
Direct segmentation (3D FCN) (Shen et al., 2019)	Full image	$512 \times 512 \times 256$	80.58	28.6656	0.8503
Patch segmentation (3D U-net) (Huang et al., 2018 ; Chen et al., 2019)	Patch	$64 \times 64 \times 64$	72.01	40.9693	3.0686
Tree data based segmentation (3D TreeConvGRU) (Kong et al., 2020)	Tree	$N \times 16 \times 16 \times 4$	68.78	30.3350	1.4336
Graph based segmentation (GCN) (Wolterink et al., 2019)	Graph	$N \times 32$	70.61	27.8718	1.2433
Baseline method (3D U-net and U-net++)	Patch	$128 \times 128 \times 64, 16^3, 32^3, 64^3$	82.96	27.2169	0.8180

They also attempted to compare previous results achieved by others, but found that due to the different datasets used (both of which were small in size), and the fact that many papers did not end up trying to reproduce other models, instead simply using the reported dice score,

meant that comparisons of previous papers was not particularly fruitful. To the best of our knowledge, no work has compared the latest CNN and ViT models with one another in the context of coronary artery segmentation in Cardiac CTs.

Ethical and clinical considerations

While developing this AI model may prove to be highly beneficial to the medicinal world, we must also think of the effects it has on the regulations and healthcare. In healthcare, AI faces serious limitations in generalisability, due to models being trained on specific data sets ([Kelly, C.J. et al.](#)). The model being trained on specific data sets prevents detection in a variety of patients potentially leading to misdiagnosis. Unlike human professionals, AI lacks contextual judgement, making it crucial to have clinical supervision with its outputs. Overreliance on automated systems without proper oversight may result in diagnostic errors compromising patient safety. Another concern is privacy and confidentiality issues, with sensitive data such as health information it is important that it is handled safely and securely. With all these risks a major component is deciding liability concerns, with the potential for misdiagnosis or unreliable outputs could require redefining standards and adjusting legal definitions of negligence and malpractice ([Cestonaro, C. et al.](#)). There are numerous considerations with incorporating AI into the healthcare industry however, we propose that this paper should be proof that ai should be used as an assistive resource to streamline and improve rather than a replacement to current methods.

Methods

ImageCAS Dataset. The dataset contains 1000 CTA 3D images of dimensions 512 x 512 x (206 – 275) voxels, far more than previously publicly available datasets. The scans imaged the heart and surrounding arteries and vessels. We used 100 images due to computational limitations. This allowed us to reduce runtime, and the computational power required to train the models, ensuring we did not exceed Kaggle’s (free resource for model training) 30-hour limit. We then split them into a training and validation set using an 80:20 split. Furthermore, the final model performance will be evaluated on a testing set that will not be included in the training or validation data sets. The dataset also contains 3D segmentation labels of the coronary arteries around the heart that correspond to each scan. This dataset was proposed by [Zeng, A et al.](#)

Preprocessing. The CTA scans were pre-processed to ensure that the dimensions were the same for each scan. Spatial padding was added to ensure each image had at least 96 x 96 x 96 dimensions. Resampling to isotropic spacing of $1 \times 1 \times 1 \text{mm}^3$ was also used to standardize the voxel spacing and the intensity was also normalized. Each image in the training set was transformed by random crop by positive negative labels to address class imbalance issues to ensure that the training is as accurate as possible.

Models

We conducted a large performance analysis between nine different segmentation models with varying sets of hyper-parameters. We grouped the models into transformer-based and convolutional-based architectures, as we aimed to identify the most accurate model architecture. Initially, we explored a mix of transformer and CNN based segmentation models. We used one channel that predicted the 3D coronary artery segmentation mask. Our model input size is $96 \times 96 \times 96$.

CNN Based Models.

UNet: Model is based on CNN architecture. UNet is composed of two main components: a contracting path and an expanding path. The contracting path aims to decrease the spatial dimensions of the image and capturing relevant information about the image by using convolution and pooling layers. The expanding path aims to up sample the feature map and produce a relevant segmentation map. This model input consists of only one channel (likely grayscale).⁵

The expanding path uses both convolution and up-convolution operations to combine learnt features and up-sample the input feature map until it creates a segmentation map. The number of channels is decreased in each block, whilst the dimensions of the matrix increase. Eventually, this will lead to a final image with the same dimensions as the input and a binary classification output.⁶

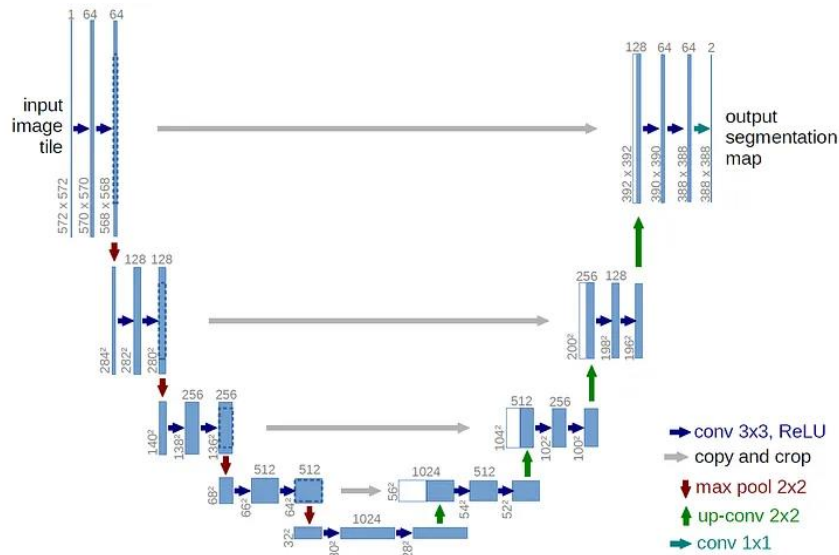


Fig 1: Architecture of a UNet model

⁵ Ronneberger, O., Fischer, P. and Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. [online] arXiv.org. Available at: <https://arxiv.org/abs/1505.04597>.

⁶ Bharath K (2024). *U-Net Architecture For Image Segmentation*. [online] Digitalocean.com. Available at: <https://www.digitalocean.com/community/tutorials/unet-architecture-image-segmentation> [Accessed 17 Apr. 2025].

SegResNet. CNN-based encoder-decoder architecture that uses a straightforward but effective design for 3D segmentation images. The model is based on the ResNet architecture which contains UNet-like encoding and decoding blocks that includes skip connections to preserve spatial information.⁷

SegResNetVAE. A model based on the SegResNet architecture; however, it also generates new data in the form of variations of the input data to increase the amount of data the model has to train with. This method avoids over fitting to the data.

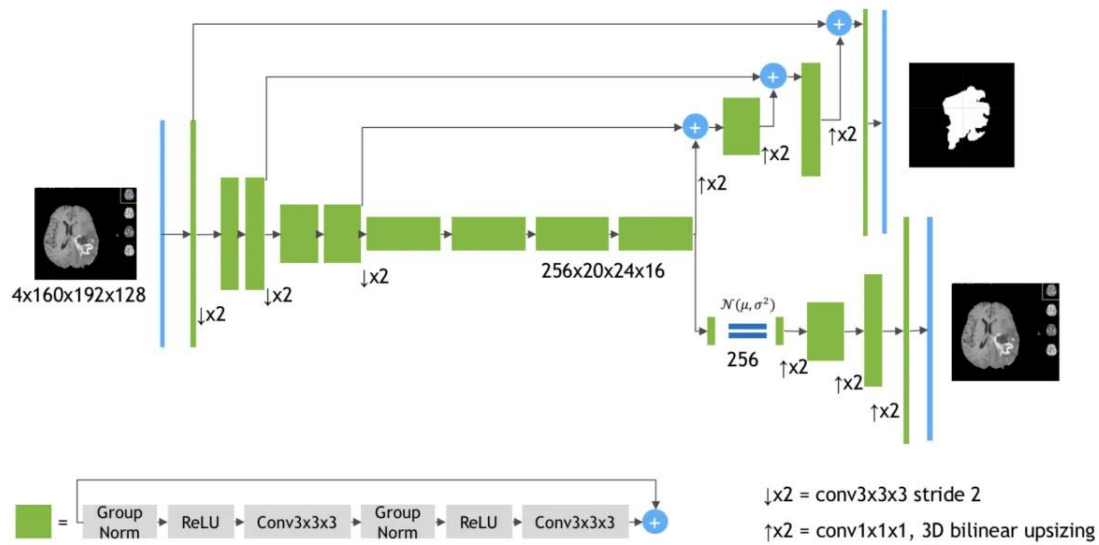


Fig 2. Architecture of a SegResNet model

DynUNet. It is an optimised version of UNet designed for medical imaging segmentation, especially in 3D imaging. It dynamically adapts to different input sizes and resolutions.

The DynUNet is a UNet-based architecture adapted from the nnU-Net framework for foetal brain segmentation in MRI. Designed to improve segmentation accuracy for super-resolution reconstruction pipelines, DynUNet adopts the nnU-Net's heuristic rules to automatically select optimal network configurations and hyperparameters. Trained on 2D patches with a combined Dice and Cross-Entropy loss, the model is validated using a 3D Dice score. Ranzini et al.⁸ demonstrate that DynUNet achieves higher performance in a single step compared to multi-step approaches, reducing the need for manual corrections in automated pipelines.

⁷ He, K., Zhang, X., Ren, S. and Sun, J. (2015). *Deep Residual Learning for Image Recognition*. [online] arXiv.org. Available at: <https://arxiv.org/abs/1512.03385>.

⁸ Ranzini, M., Fidon, L., Ourselin, S., Modat, M. and Vercauteren, T., 2021. MONAifbs: MONAI-based fetal brain MRI deep learning segmentation. *arXiv preprint arXiv:2103.13314*. <https://arxiv.org/pdf/2103.13314>

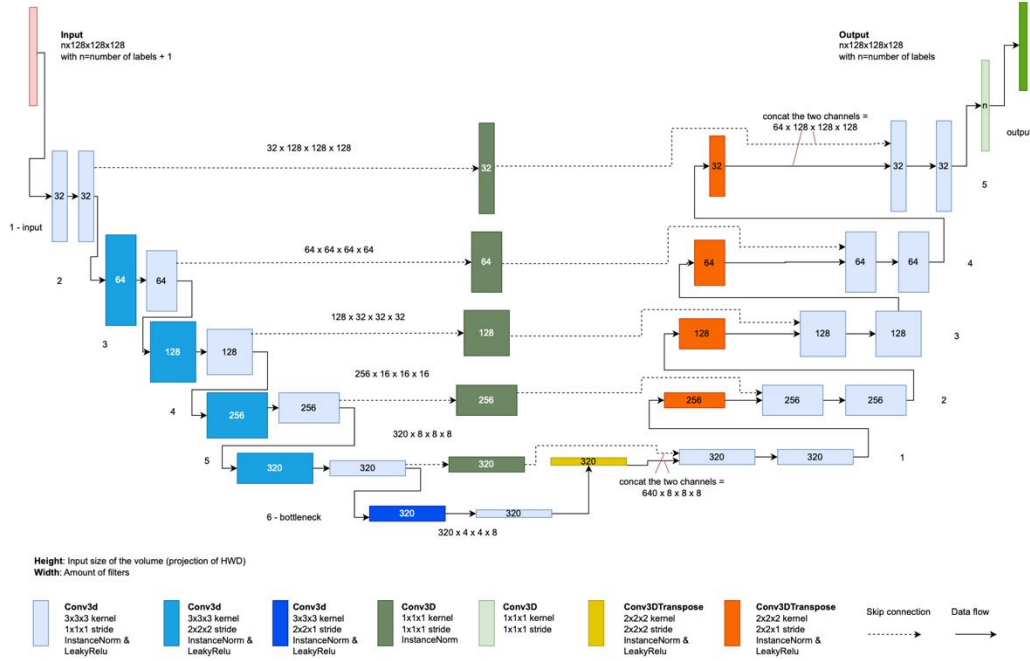


Fig 3. Architecture of a DynUNet model

VNet. This Model is a modified version of Unet specifically for 3D segmentation.^{9 10} The left part of the network consists of a compression path and the right part decompresses the signal until its original size is reached. The compression branch is divided in different stages that contain between one to three convolutional layers. PReLU non linearities are applied throughout the network. This function improves upon Leaky ReLU by making the slope a learnable parameter which enhances model accuracy and convergence. However, fine-tuning this parameter can be more time-consuming.

The right portion of the network extracts features and expands the spatial support of the lower resolution feature maps. The feature maps computed by the very last convolutional layer have $1 \times 1 \times 1$ kernel size, producing outputs of the same size as the input volume that are converted to probabilistic segmentations of the foreground and background regions by applying soft-max voxel wise. Our program rounds the probability of each pixel to 0 and 1 and switches each pixel on and off accordingly to generate the segmentation label.

⁹ arSiv. (2015). *V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation*. [online] Available at: <https://arSiv.labs.arxiv.org/html/1606.04797> [Accessed 17 Apr. 2025].

¹⁰ NVIDIA NGC Catalog. (2023). *V-Net Medical for TensorFlow | NVIDIA NGC*. [online] Available at: https://catalog.ngc.nvidia.com/orgs/nvidia/resources/vnet_medical_for_tensorflow [Accessed 17 Apr. 2025].

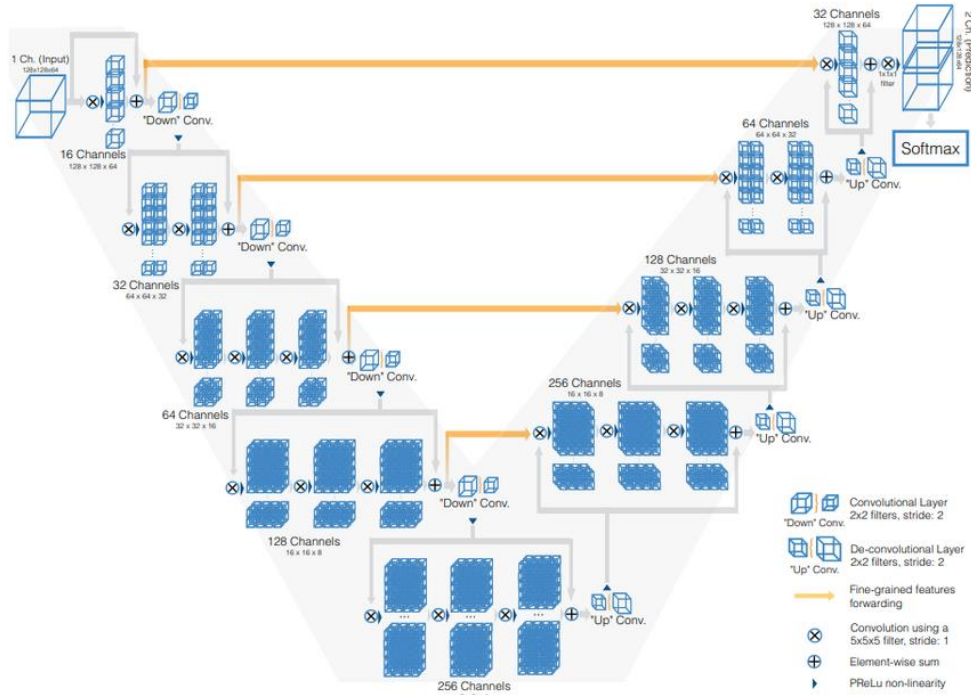


Fig 4. Architecture of a SegResNet model

BasicUNetPlusPlus. This model introduces dense skip connections and nested convolution blocks to enhance feature learning and segmentation accuracy. While UNet has basic feature propagation this model has enhanced hierarchal feature fusion which allows for more precise object boundaries, better multi-scale learning and reduced information loss, stronger gradient flow and improved performance on small and complex objects.¹¹

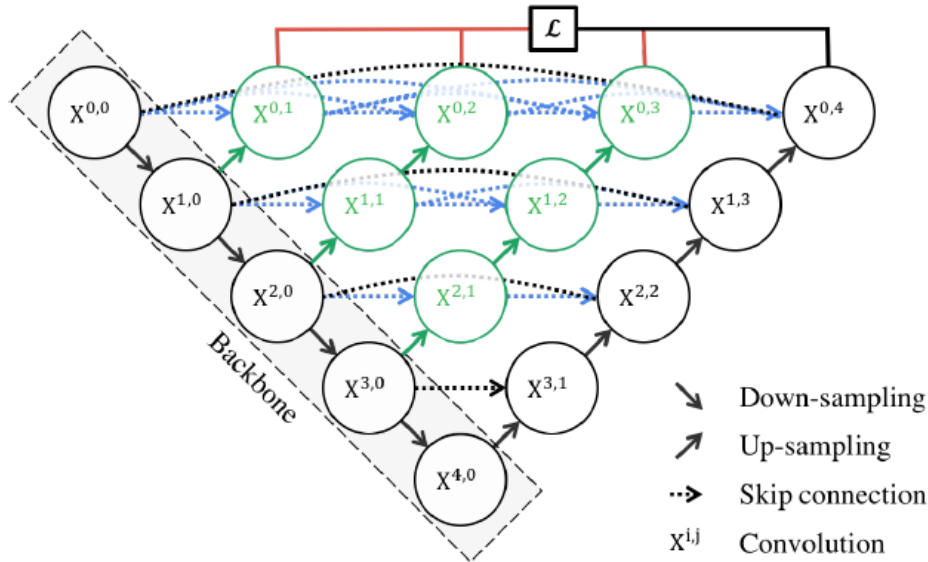


Fig 5.

¹¹ Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N. and Liang, J. (2018). UNet++: A Nested U-Net Architecture for Medical Image Segmentation. *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*, [online] 11045, pp.3–11. doi:https://doi.org/10.1007/978-3-030-00889-5_1.

Transformer Based Models.

SwinUnetR. The architecture incorporates transformers and MLP's. The model uses similar encoder-decoder structure of U-Net however the model replaces the classic convolutional encoder with a hierarchical 3D Swin Transformer. The encoder gradually compresses the input image into feature representations, and the decoder up samples these features to produce a segmentation map. The model uses attention blocks as its encoder instead of solely CNN. SwinUnetR¹²¹³

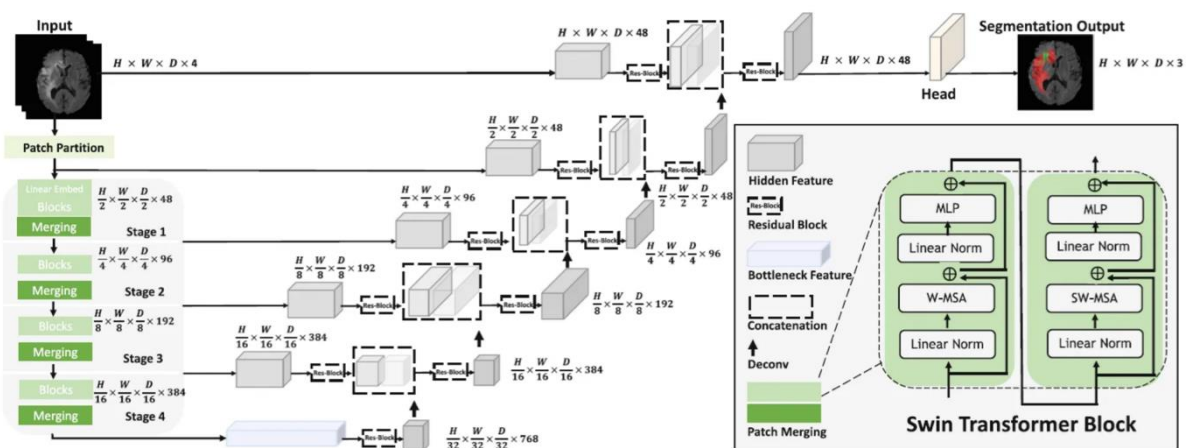


Fig.6. The SwinUnetR network

AttentionUnet. This model is based on the UNet structure with additional attention blocks. During up-sampling in the expanding path, spatial information recreated is generally imprecise. To counteract this problem, the UNet uses skip connections that combine spatial information from the down sampling path with the up-sampling path. However, this brings across many redundant low-level feature extractions, as feature representation is poor in the initial layers.

¹³Hatamizadeh, A., Nath, V., Tang, Y., Yang, D., Roth, H. and Xu, D. (2022). Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. *arXiv:2201.01266 [cs, eess]*. [online] Available at: <https://arxiv.org/abs/2201.01266>.

AttentionUnet implements soft attention at the skip connections that actively suppresses activations in irrelevant regions, reducing the number of redundant features brought across the skip connections.¹⁴

The attention gate seen below, takes two vectors as input, g and x . The x input comes from a lower down layer of the network which feeds in smaller dimensions and better feature representation. The smaller vector x , goes through a strided convolution such that its dimensions match vector g . The two vectors are added up, passed through a ReLU. This sum then goes through a sigmoid layer which scales the vector between the range $[0,1]$, producing the attention coefficients, where coefficients closer to 1 indicate more relevant features to pass through the skip connection where this vector (containing 0 and 1) is multiplied by the up sampled image to remove any redundant features when multiplied by number close to 0, and keeping features when multiplied by numbers close to 1.¹⁵

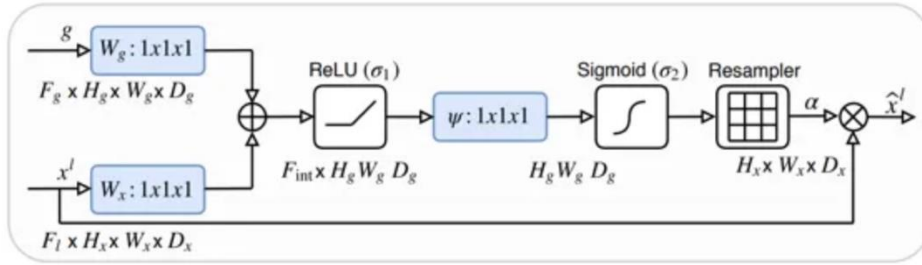


Fig 7.

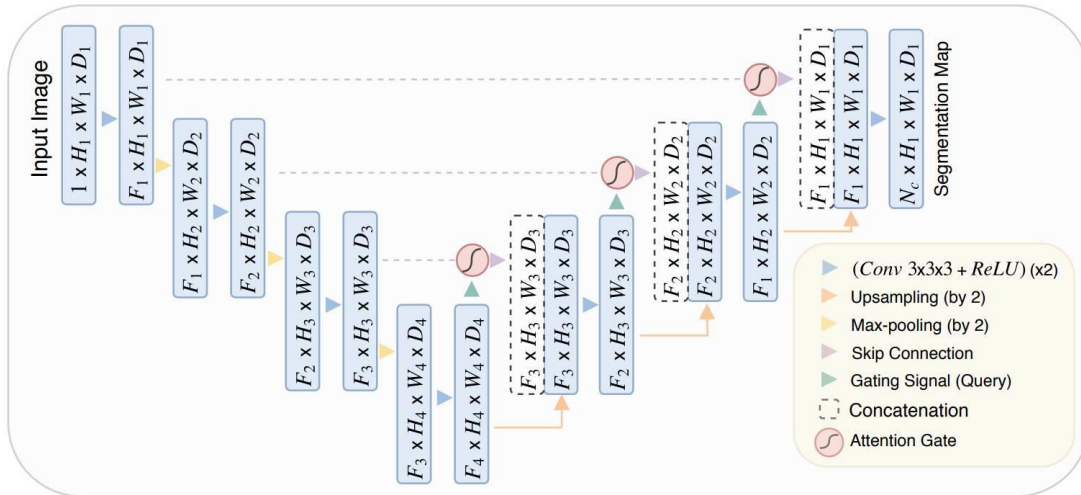


Fig 8. Architecture of a AttentionUNet model

¹⁴ Oktay, O., Schlemper, J., Folgoc, Loic Le, Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, Nils Y, Kainz, B., Glocker, B. and Rueckert, D. (2018). *Attention U-Net: Learning Where to Look for the Pancreas*. [online] arXiv.org. Available at: <https://arxiv.org/abs/1804.03999>.

¹⁵ Tomar, N. (2023). *Attention UNET and its Implementation in TensorFlow - Idiot Developer*. [online] Idiot Developer. Available at: <https://idiotdeveloper.com/attention-unet-and-its-implementation-in-tensorflow/> [Accessed 17 Apr. 2025].

UnetR. UnetR is a transformer-based architecture which integrates transformer-based encoders with the traditional U-net framework. It works by dividing the image into on-overlapping patches, each patch is flattened and linearly embedded into a sequence forming the input for the transformer encoder. The embedded patches are processed by the transformer encoder, which captures the long-range dependencies and global contextual information across the volume. This allows the model to understand complex anatomical structures and their relationship within the 3D spaces. The decoder reconstructs the segmentation map by up sampling and combining features received from the encoder and skip connection, this step-by-step up sampling helps in generating high-resolution segmentation outputs. Transformers excel at capturing long range dependencies within coronary artery segmentation, detailed feature extraction and improve handling of complex structures.¹⁶

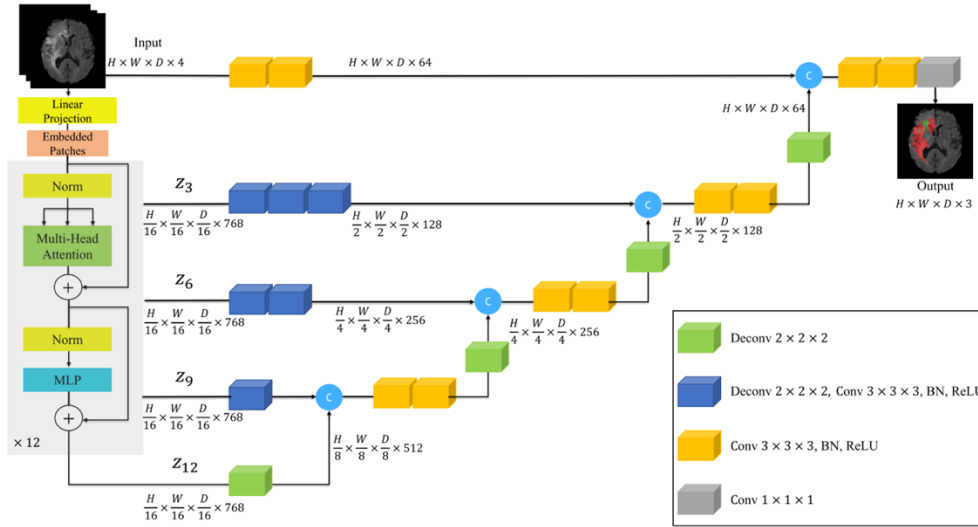


Fig 9. Architecture of a UNetR model

Experimental Setup

Our model was build using PyTorch on the Kaggle platform where we used the P100 GPU accelerator.

Hyper parameters

Parameter	Value
Roi	96 x 96 x 96
Batch Size	1
Max Epochs	10
In Channels	1
Out Channels	1

¹⁶ Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H. and Xu, D. (2021). UNETR: Transformers for 3D Medical Image Segmentation. *arXiv:2103.10504 [cs, eess]*. [online] Available at: <https://arxiv.org/abs/2103.10504>.

Spatial Dims	3
Learning Rate	0.001
Weight Decay Rate	0.00001

Metric (dice formula)

$$DSC = \frac{2 |X \cap Y|}{|X| + |Y|}$$

Our main metric of accuracy was the dice metric that compares the overlapping area between the predicted segmentation and the training label. This formula yields a result between 0 and 1 indicating a perfect match. The best scoring models achieved scores of c. 0.7, which is a 70% overlap. Our formula does add a very small value to the denominator to avoid dividing by 0 on labels that do not overlap.

The DiceLoss is also an indicator of whether the model is still improving or if it has plateaued, by tracking this, we could stop training the model when the loss does not reduce anymore to avoid over fitting the model to our small data set. The loss is also used to optimize the model parameters.

Optimizer

For low epoch training cycles, we used the AdamW optimisation function. This function combines the well-known Adam optimisation function and decoupled weight decay. Our learning rates were 0.001 and our weight decay was 0.0001. We included the weight decay to prevent over fitting on our set of 100 3D images. For larger test runs, we implemented a scheduler that used the cosine annealing function. The scheduler modifies our learning rate in the below pattern (Fig 10). The benefit of constantly changing the learning rate is so that the gradient descent function can find the lowest minima by allowing the gradient to take larger

steps to find lower minima to reduce the loss function and increase the dice score which would ultimately increase the model's accuracy and real-world performance.

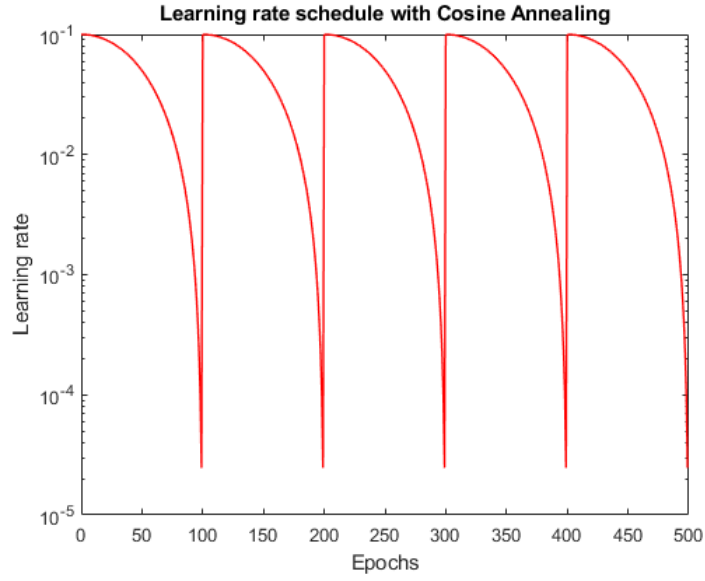


Fig. 10

Results

Model	Hyperparameter	Dice Metric	Dice and Loss Image	3D Predicted Image
UNetR (Transformer)	img_size = 96x96x96	0.654	Fig. 11	Fig. 12
AttentionUNet (CNN)	channels = (16, 32, 64, 128, 256) strides = (2, 2, 2, 2)	0.225	Fig. 22	
SwinUNetR (Transformer)	feature_size = 24	0.720	Fig. 13	Fig. 14
DynUNet (CNN)	kernel_size = (3, 3, 3, 3) strides = (1, 2, 2, 2) upsample_kernel_size = (2, 2, 2) filters = (16, 32, 64, 128)	0.731	Fig. 15	Fig. 16
SegResNet (CNN)	init_filters = 16	0.659	Fig. 17	Fig. 18
UNet (CNN)	channels = (16, 32, 64, 128, 256), strides = (2, 2, 2, 2) num_res_units = 2	0.591	Fig. 19	Fig. 20
VNet (CNN)	spatial_dims = 3	0.161	Fig. 21	
SegResNetVAE (CNN)	init_filters = 8 use_conv_final = True	0.697	Fig. 23	

We also trained one SegResNet model for 20 epochs and 200 3D CTA images that trained for c. 7 hours on Kaggle. It scored a dice metric of 0.760 which proves that adding more images and epochs improves model accuracy at the cost of increased computational power requirements. The graphs of loss and dice metric for this model are shown in figure 24. This example shows that increasing the training time will increase the model accuracy. In real world examples models should be trained on the full 1000 3D CTA scans.

Discussion

Out of all the deep learning-based models, DynUNet (CNN based) achieved the highest dice score, at 0.731, with SwinUNetR (transformer based), right behind at 0.720. Although DynUNet did achieve the highest dice, the mask it predicted ended up with a lot of noise, which may suggest that it is less suitable. On the other end, VNet and AttentionUNet performed substantially worse compared to the rest of the models, both with very low dice scores (0.161 and 0.225 respectively). These results are not in line with what has been achieved by others before, with VNet having had a dice score of 0.869 ± 0.033 ([Ahmadi, S., Milletari, F., Navab, N.](#)) and AttentionUNet 0.943 ± 0.092 ([Oktay, O. et al.](#)). A possible reason for these discrepancies is the difference in what is being imaged. The scores above were reflections of the segmentation of the prostate and spleen respectively, both using different datasets, making a direct comparison difficult, and perhaps suggesting that these models are best suited for segmenting organs rather than arteries. Similar arguments apply to comparisons between most models shown. Also, the low number of epochs used to train the models most likely affected the final dice score of these two models.

We had issues with the limitations of the Kaggle GPUs (P100) and their limited memory. We were unable to train very complex models, and we could not use all 1000 images. We also had to cut down the number of epochs which harmed model accuracy. However, we ensured that all models had hyperparameters that are at a similar level of complexity to avoid some being better than others. On average each model took 2 hours to train suggesting the models had similar levels of compute required to train. Models used in the real world will have access to larger GPU clusters and 3D images.

Some improvements can be made to the model such as post processing the final segmentations to cut out the labels which do not have similar label areas above and below the layer, this can remove areas where the model did not predict the artery correctly but may have highlighted random noise. Using all 1000 3D CTA scans will increase model accuracy and could show dice scores closer to 0.9. Another improvement is by increasing the complexity of the model architecture so that it can extract more information from the scan. We limited the number of epochs used to train the model at 10 so we did not use the scheduler optimiser. When the number of epochs used in real world increases, the scheduler should be able to find lower minima with cosine annealing and therefore yield more accurate results.

Conclusion

Segmenting the coronary artery from a CTA scan manually to detect CHD, the leading cause of death in the world, is a tedious and error prone process. In this paper, we have compared the efficacy of CNN and vision transformer based deep learning models for automating this segmentation, finding that overall, CNNs were just barely outperformed the vision transformer-based models, using the dice score as our metric for comparison. We used a dataset of 100 images, and trained each model for 10 epochs, and found that the CNN based DynUNet performed best in this scenario, with a dice score of 0.731, just ahead of SwinUNetR (vision transformer based) at 0.720. However, due to noise in the predicted mask, it is hard to definitively call DynUNet the best model for this scenario. It is very possible that deep learning models are applied in all labour intensive fields in the medical world due to their speed and efficacy when trained with large enough datasets and computing resources.

Supplementary

When we coded our model, we had to iterate over many different variations until we found the best structure. For example, one main issue relating to the accuracy of the model was how we transformed the 3D images for the training data set. As a precautionary step we applied spatial padding to all dimensions of the 3D image to ensure equal sized tensors. An intensity normaliser was applied to make all voxel values between 0 and 1. For example, we made changes to figure 26, to arrive at the code shown in figure 25. The transformations in figure 25 yielded much better results than the transformations in figure 26. We improved the code in figure 26 by adding the intensity normaliser and the voxel-spacing resampling. The normalisation layer scales each image into a roughly consistent range between 1 and 0. In figure 26 that step is gone, so the network is trying to learn from highly varying values (as different scanners, different patients and arbitrary Hounsfield-unit ranges change the expected output values making it much harder for the model to learn). Without the voxel-spacing resampling every 3D image will have different inter-slice thickness and in-plane resolution. For example, a 2 mm slice thickness CT and a 5 mm one look very different to a network, so features won't line up spatially and performance collapses.

```
transforms.LoadImaged(keys=["image", "label"], ensure_channel_first=True),
transforms.SpatialPadd(keys=["image", "label"], spatial_size=roi),
transforms.NormalizeIntensityd(keys="image", nonzero=True, channel_wise=True),
transforms.Spacingd(keys=["image", "label"], pixdim=(1.0, 1.0, 1.0), mode=("bilinear", "nearest")),
transforms.RandCropByPosNegLabeld(keys=["image", "label"],
```

Fig. 25

```

transforms.LoadImaged(keys=["image", "label"], ensure_channel_first=True),
transforms.SpatialPadd(keys=["image", "label"], spatial_size=roi),
transforms.RandCropByPosNegLabeld(keys=["image", "label"],
                                   label_key="label",
                                   spatial_size=roi,
                                   num_samples=num_crop_samples,
                                   image_key="image", ),

```

Fig. 26

Link to our public Kaggle code file:

<https://www.kaggle.com/code/zackzlatarev123/sparc>

Project Organisation

We held meetings with our supervisor every week learning about the key concepts of machine learning, asking any questions we had about the code and any advice he may have for our project, as well as spending a few hours a week adding to the notebook, whether it be training different models, fixing problems in code, or changing different transforms applied to the dataset

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Images:

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Images for Results

UNnetR

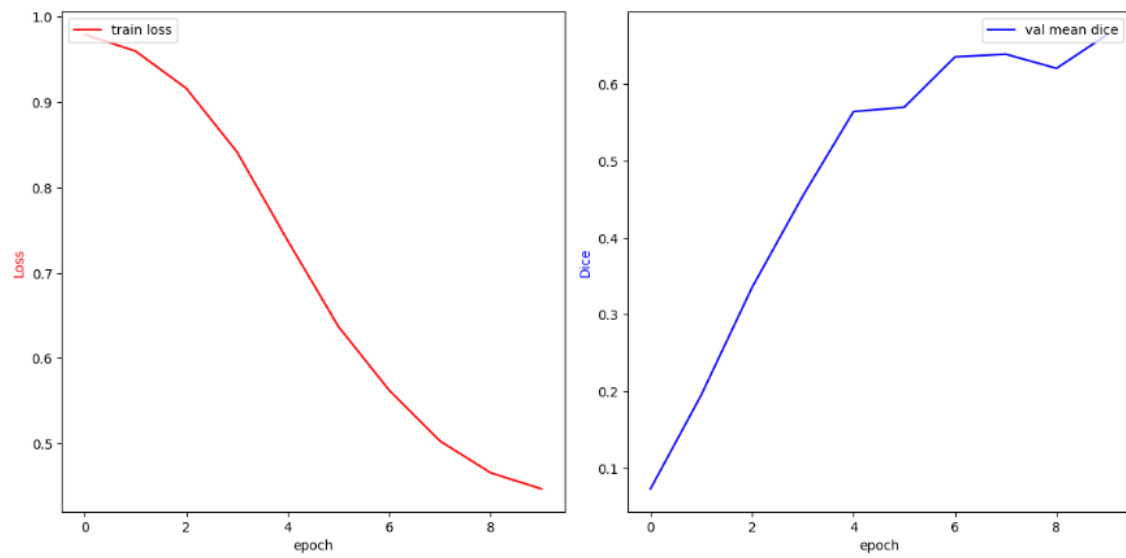


Fig. 11

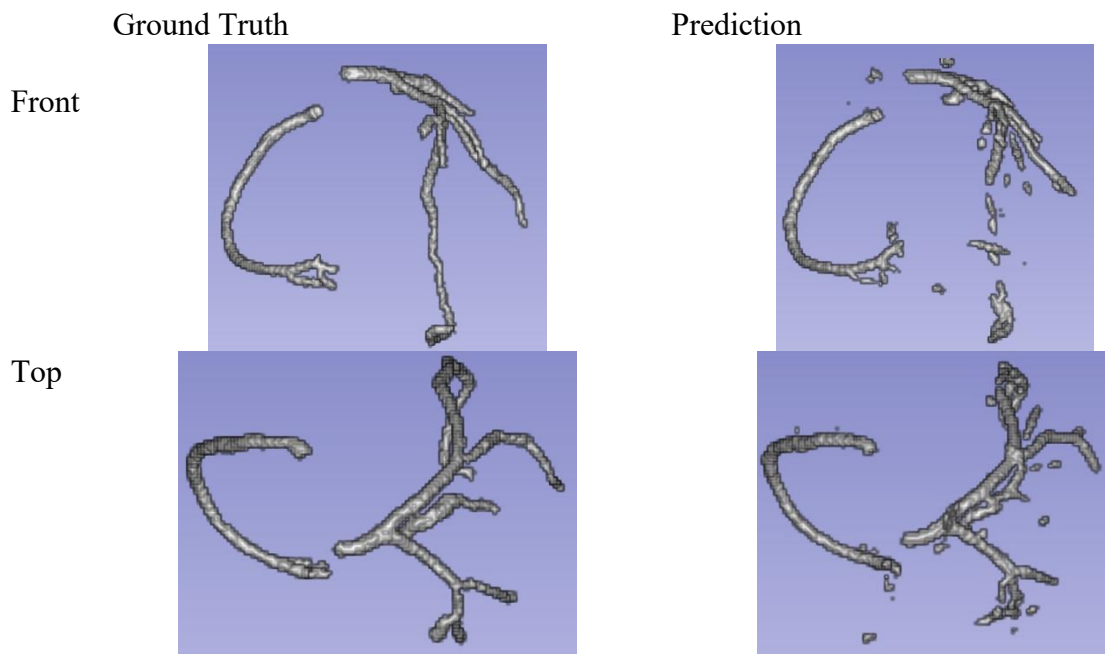


Fig. 12

SwinUnetR

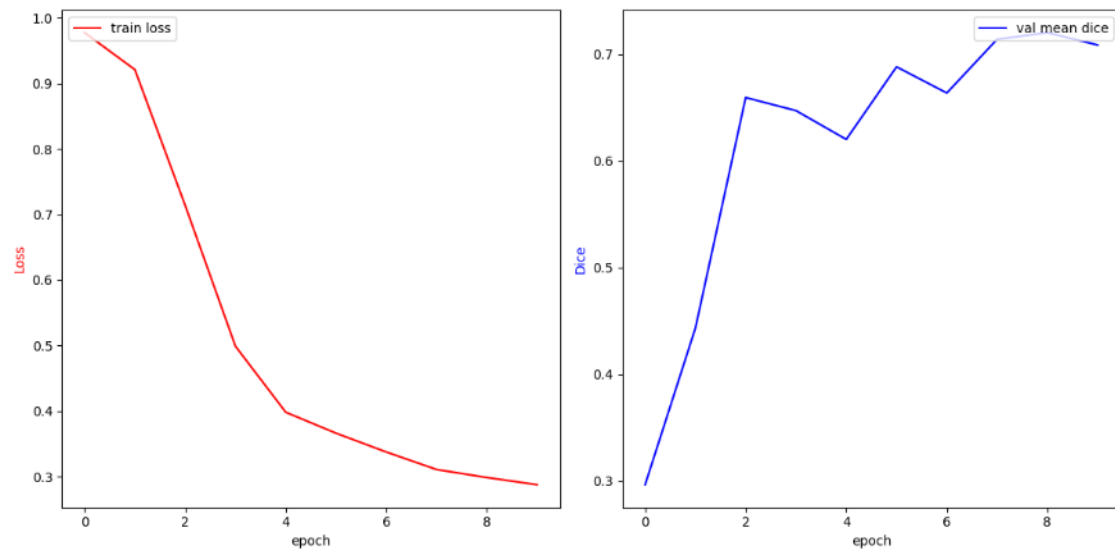


Fig. 13

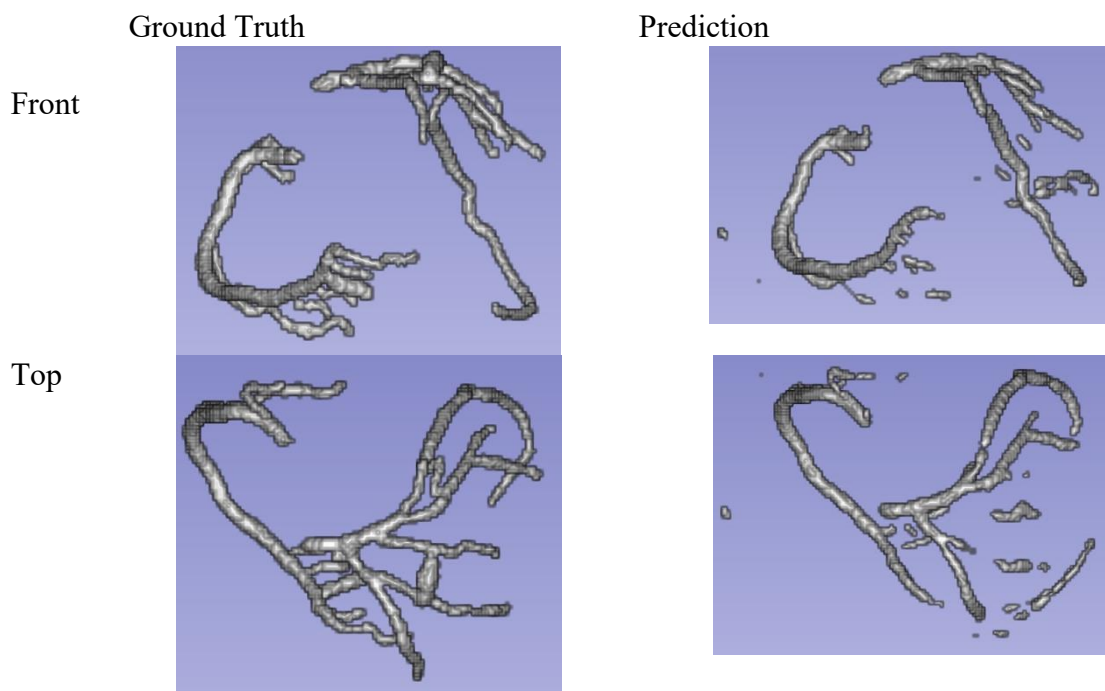
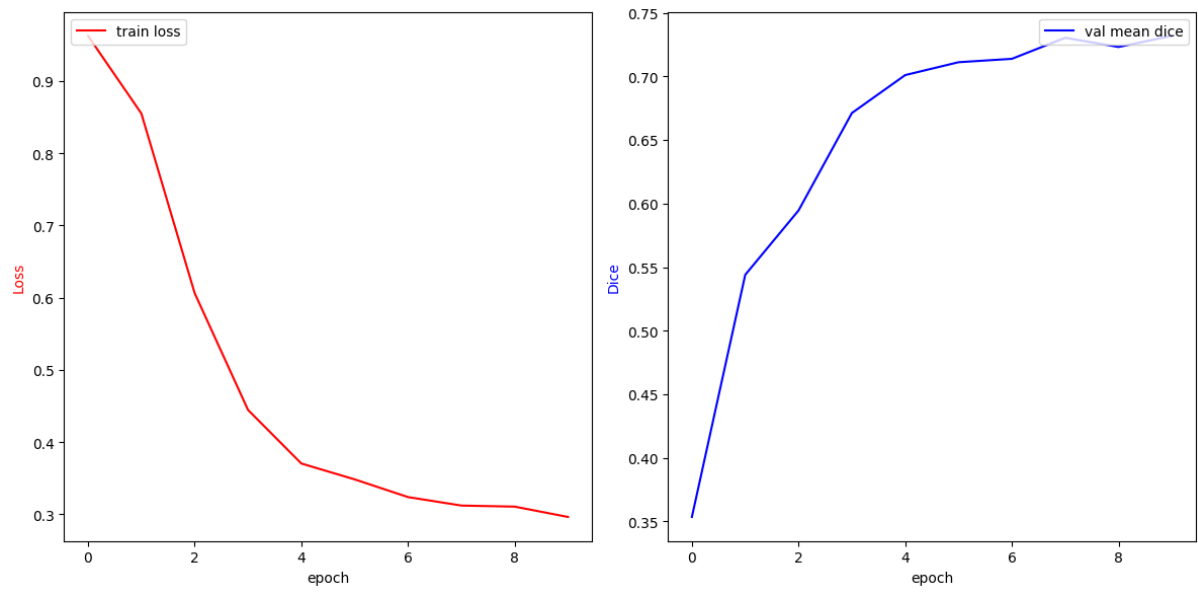


Fig 14

DynUnet



SFig. 15

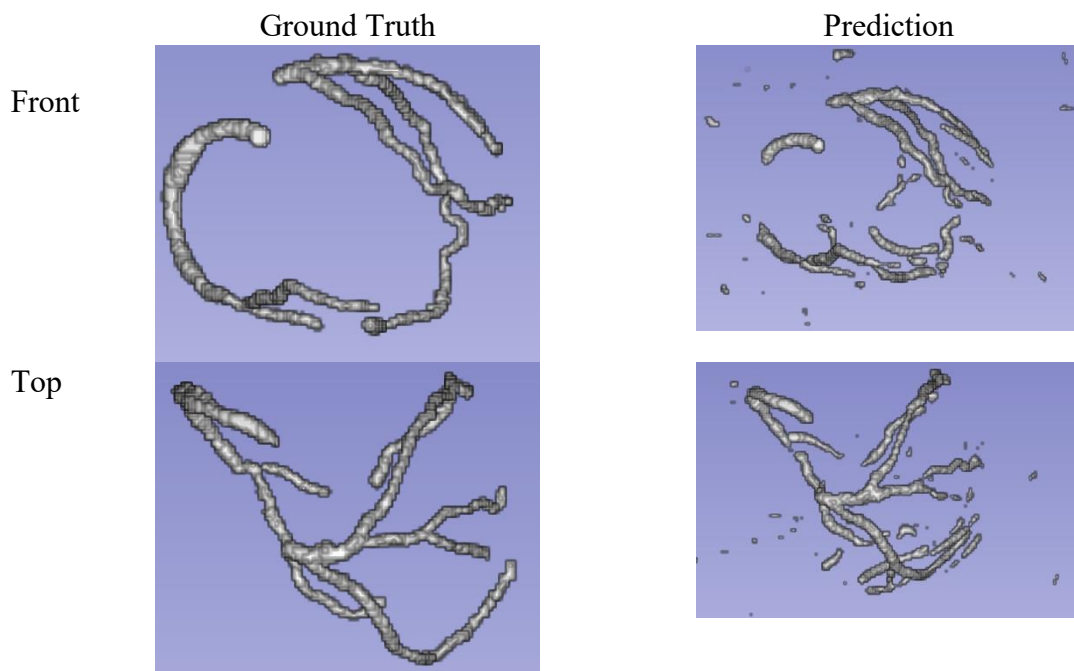


Fig 16

SegResNet

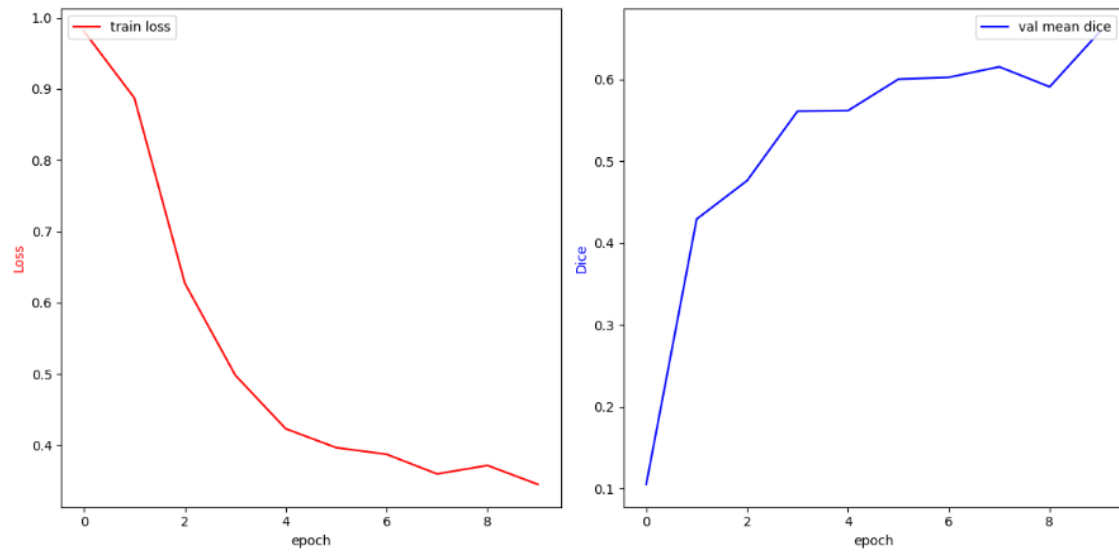


Fig. 17

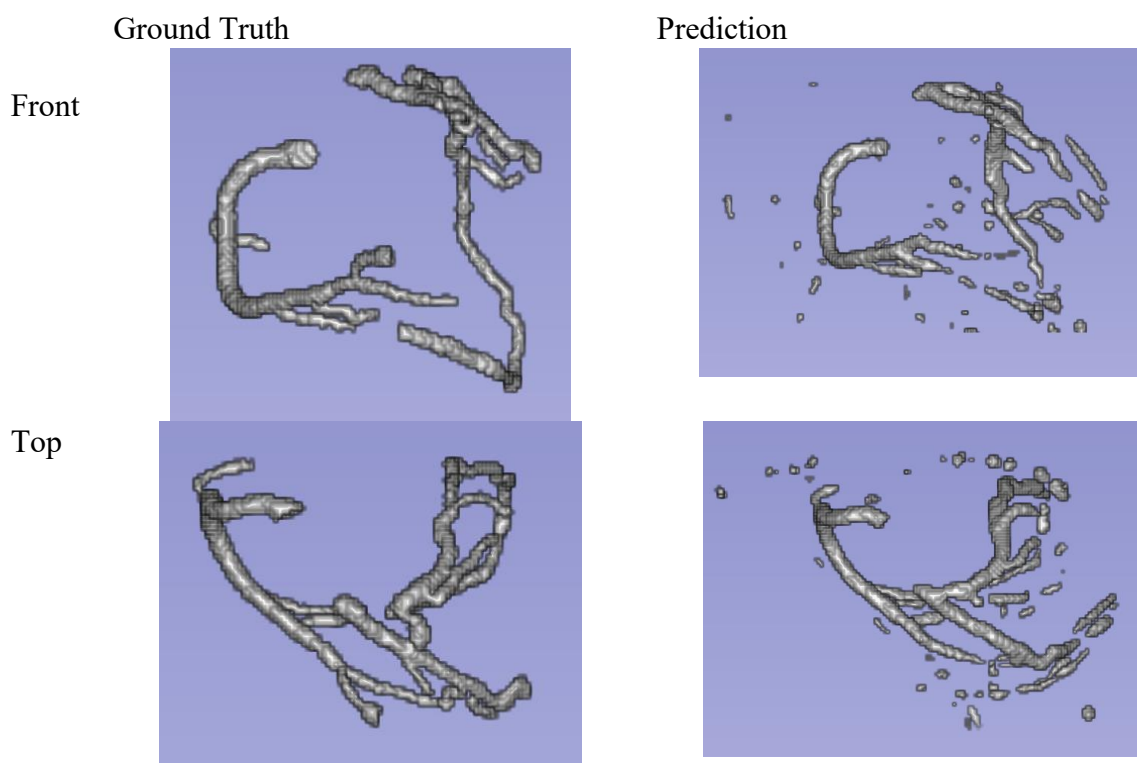


Fig. 18

Unet:

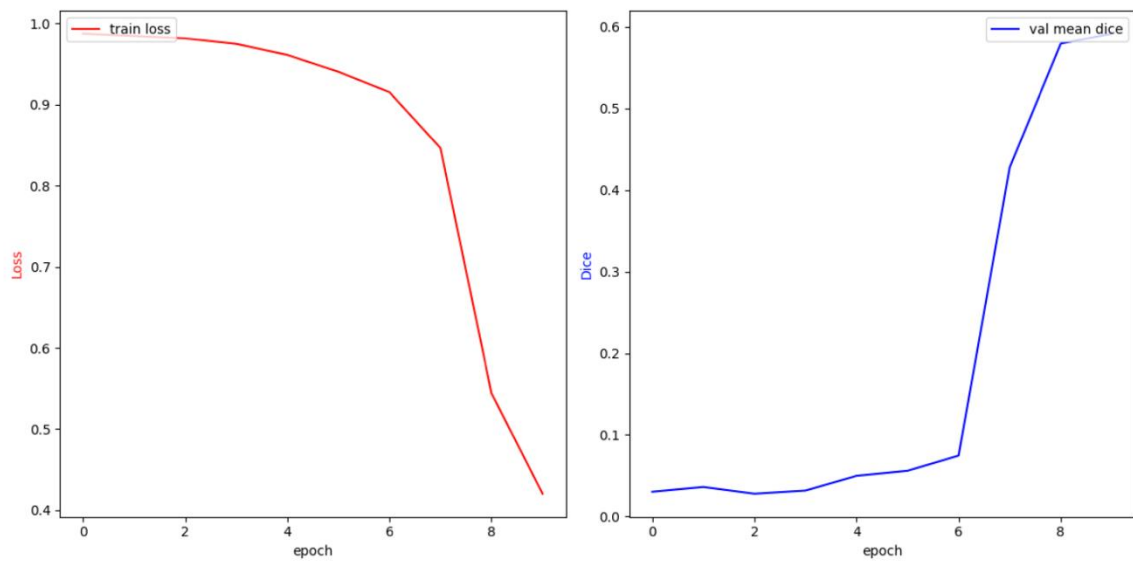


Fig. 19

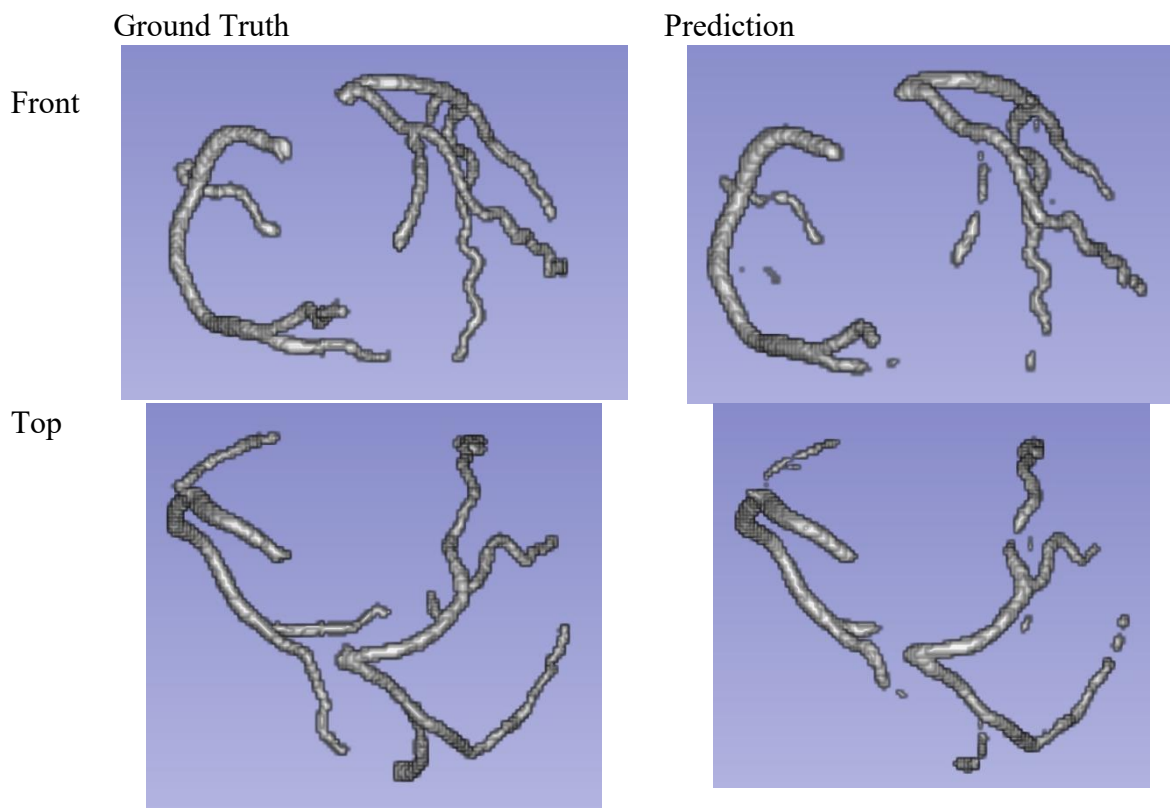


Fig. 20

VNet:

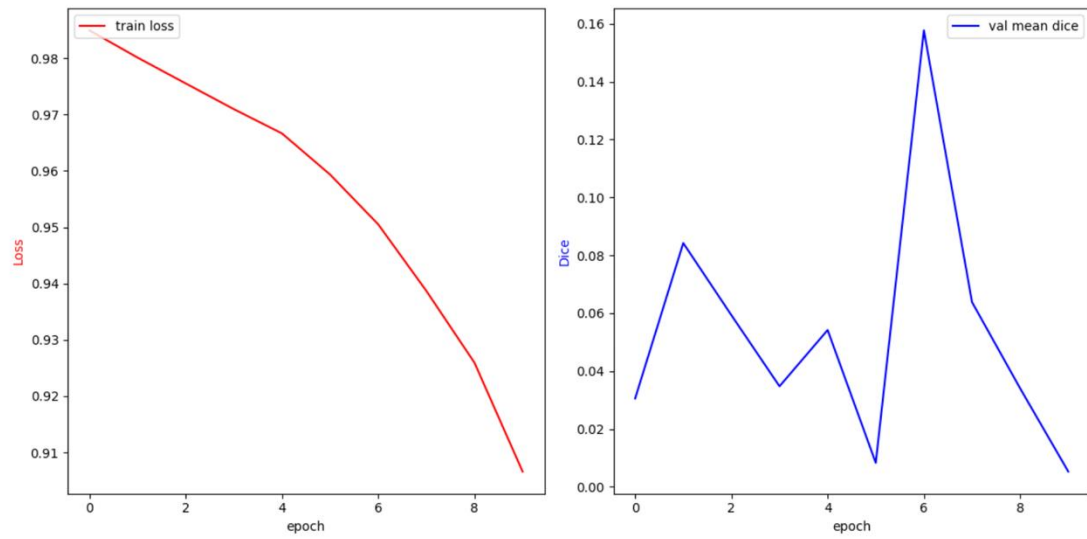


Fig. 21

AttentionUnet

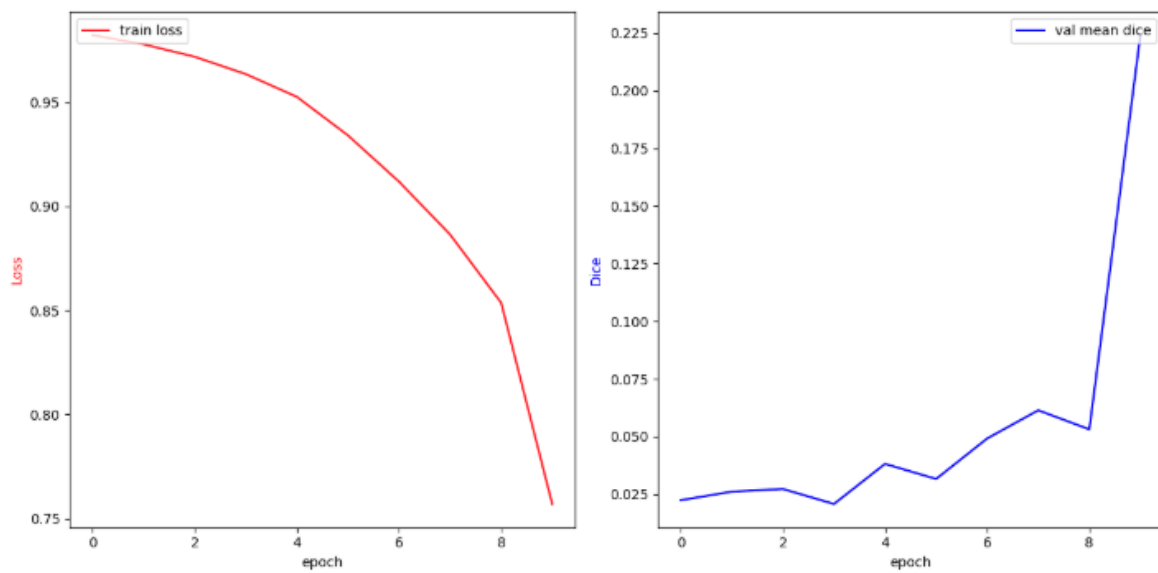


Fig. 22

SegResNetVAE:

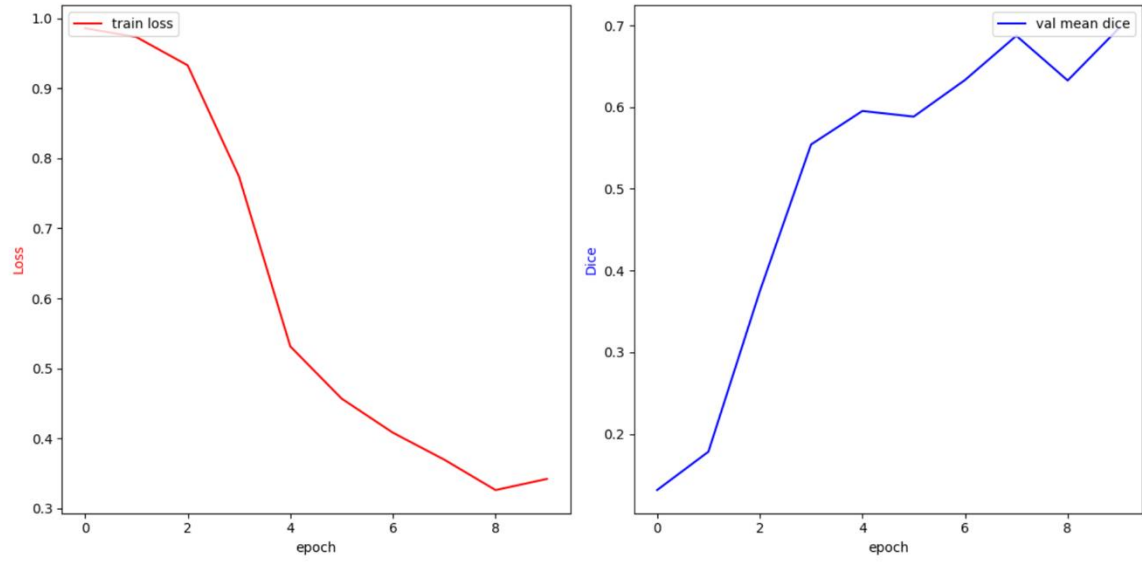


Fig. 23

SwinUNETR (200 images, 20 Epochs)

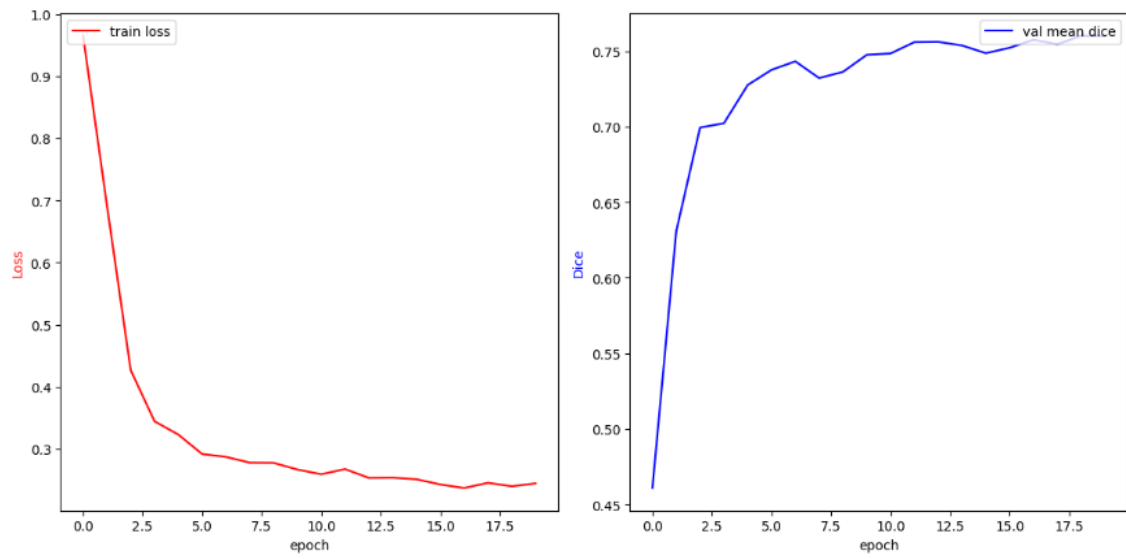


Fig. 24

Loss function

$$\mathcal{L}(G,Y)=1-\frac{2}{J}\sum_{j=1}^J\frac{\sum_{i=1}^I G_{i,j}Y_{i,j}}{\sum_{i=1}^I G_{i,j}^2+\sum_{i=1}^I Y_{i,j}^2}-\frac{1}{I}\sum_{i=1}^I\sum_{j=1}^J G_{i,j}\log Y_{i,j}.$$

Dice score metric

$$\text{Dice}(G, P) = \frac{2 \sum_{i=1}^I G_i P_i}{\sum_{i=1}^I G_i + \sum_{i=1}^I P_i},$$

$$\text{HD}(G', P') = \max \left\{ \max_{g' \in G'} \min_{p' \in P'} \|g' - p'\|, \right. \\ \left. \max_{p' \in P'} \min_{g' \in G'} \|p' - g'\| \right\}.$$